

DSCI445: Midterm Review

Fall 2026

Model Assessment

1. I hope your speech in Ohio went well last week and that you had a nice weekend. I wanted to reach out with a question/something I was hoping you could go over in more detail during the review session. I was hoping you could go over how to use cross-validation in classification problems and how we can apply it to optimize figuring out the best K for KNN.
2. I am still a little confused about k -fold cross-validation for classification, specifically when we are using KNN. I understand the general idea, but I am not entirely sure what it means to “tune” the model in this context. “Tuning” comes up multiple times in the code, and I would appreciate going over what the process is actually doing.
3. I would also like to go over the bias-variance tradeoff for the different cross-validation methods again, particularly LOOCV. I understand that LOOCV does not involve randomness in how the folds are selected, so I am confused about why LOOCV has higher variance than k -fold. My current understanding is that it has something to do with the fact that the training sets contain almost the same information entirely and that blows up the covariance, but I would really appreciate an explanation to clarify things and solidify my understanding.
4. When splitting data for the validation set or KNN or LOOCV are there any glaring warnings to be looking for, when different choice may be necessary?
5. When we are assessing the Test MSE and Train MSE page 3 of chapter 3 notes and what is best for one is worst for the other, would estimating the test MSE lead to you to choose the most similar model?

Other questions

1. I'm a little confused at how the Bayes theorem works for Linear Discriminant Analysis. Not sure about how to use it for estimating different characteristics for different predictors.
2. When working with the `ggpairs()` function it is somewhat overwhelming to identify the relationships within the plots, is there something I can run alongside this call to gain insights of the relationships, or any reminders of what to look for visually?

Questions I originally missed in my inbox

1. Choosing the “best” model sometimes feels like a mechanical switch of “which test MSE is smaller” or in general “which error is the least possible”, but what I keep wondering is what about non-parametric decisions? What about a model is the best one as a tradeoff between lowest test MSE, interpretability and computational and real-world expense? I think my question goes about whether testing which model is best can also come from a thought-process rather than a mere checking which-one-has-the-lowest-error one.
2. What about the thin difference in categorical vs quantitative in some real-world problems. If the bounds and ranges to make a quantitative variable categorical are scientifically driven and yield a more structured prediction and inference, why would sticking with the quantitative approach still be best? Is there something I am missing?
3. What if the randomization process of drawing a subset of the data to make it the validation one yields a particularly-characterized validation or training sample which model’s estimates would be odd and would be yielding particular results than if with a very different “randomized” subset? How can we be more conservative regarding non-meeting this assumption of non-randomized sub-sample draws, to still be safely in the validation/training split as if we had 2 datasets to use them for these reasons?